

GEOGRAPHY

INDIAN GEOGRAPHY



INDIAN PHYSIOGRAPHY

MODULE - 5

PHYSIOGRAPHIC DIVISIONS **THE GREAT PLAINS OF INDIA**

REGIONAL DIVISIONS OF GREAT PLAINS OF INDIA



1

Although, the Great Plain of North India is treated as a Geographical unit with low elevation and gentle slope, this vast areas exhibits distinctive fluvial patterns, and directions of flow in different parts allowing it to be divided into the following four major regions.

- The Rajasthan Plain
- The Punjab-Haryana Plain
- The Ganga Plain
- The Brahmaputra Plain

THE GREAT PLAIN OF NORTH INDIA



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REGIONAL DIVISIONS OF GREAT PLAINS OF INDIA

1

THE RAJASTHAN PLAIN



1

The western extremity of the Great Plains of India consists of the **Thar** or the **Great Indian Desert**, which covers western Rajasthan and adjoining regions of Pakistan.

2

This vast desert is an **undulating plain** dotted with **longitudinal dunes** and **barchans** whose average elevation is about 325 m above mean sea level.

3

The desert is called **Marusthali** and it accounts for the greater part of the Marwar plain

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THE RAJASTHAN PLAIN

- 4 This region receives low rainfall below 150 mm per year; hence, it has arid climate with low vegetation cover.



- 5 It is because of these characteristic features that this is also known as Marusthali.

- 6 It is believed that during the Mesozoic era, this region was under the sea.

- 7 This can be corroborated by the evidence available at wood fossils park at Aakal and marine deposits around Brahmsar, near Jaisalmer (The approximate age of the wood fossils is estimated to be 180 million years).

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THE RAJASTHAN PLAIN



8 It has a vast stretch of sand with a few **outcrops of gneisses, schists and granites** which proves that geologically it is part of the Peninsular Plateau.

9 Though the underlying rock structure of the desert is an extension of the Peninsular plateau, yet, due to extreme arid conditions, **its surface features have been carved by physical weathering and wind actions.**

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THE RAJASTHAN PLAIN



Ephemeral Rivers

- *They flow for very short periods following heavy rainfall or snowmelt.*
- *For the rest of the year, river's channel remains dry.*
- *Such rivers are generally found in arid areas specially near the tropics.*

10

Some of the well pronounced desert land features present here are mushroom rocks, shifting dunes and oasis (mostly in its southern part).

11

On the basis of the orientation, the desert can be divided into two parts: the northern part is sloping towards Sindh and the southern towards the Rann of Kachchh.

12

Most of the rivers in this region are ephemeral.

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THE RAJASTHAN PLAIN



13

In general, the eastern part of Marusthali is rocky while its western part is covered by shifting sand dunes locally known as **Dhrian**.

14

It drained by a number of short seasonal stream originating from the **Aravalis** and it supports agriculture in some patches of fertile tracts. These fertile tracts are known by the name of **Rohi**.

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THE RAJASTHAN PLAIN



15 Even the important river, the Luni, is a seasonal stream which flows towards the south-west to the Rann of Kuchchh.

16 North of the Luni basin, there is large area of inland drainage on the eastern edge of the Thar desert having several saline lakes.

17 They are source of common salt and many other salts. The largest of such lakes is the Sambhar Lake near Jaipur.

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THE RAJASTHAN PLAIN



18 Low precipitation and high evaporation makes it a **water deficit region**.

19 There are some streams which disappear after flowing for some distance and present a **typical case of inland drainage** by joining a lake or **playa**.

20 The lakes and the playas have **brackish water** which is the main source of obtaining salt.

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2

PUNJAB-HARYANA PLAIN

1 The Great Indian Desert imperceptibly gives way to the fertile land of Punjab and Haryana towards the east and north east.

2 The entire plain is formed by the alluvial deposits of five rivers:

- The Sutlej
- The Beas
- The Ravi
- The Chenab
- The Jhelum

3 This is known as the Punjab Plain-the land of five rivers.

4 It is primarily made up of doabs-the land between two rivers.

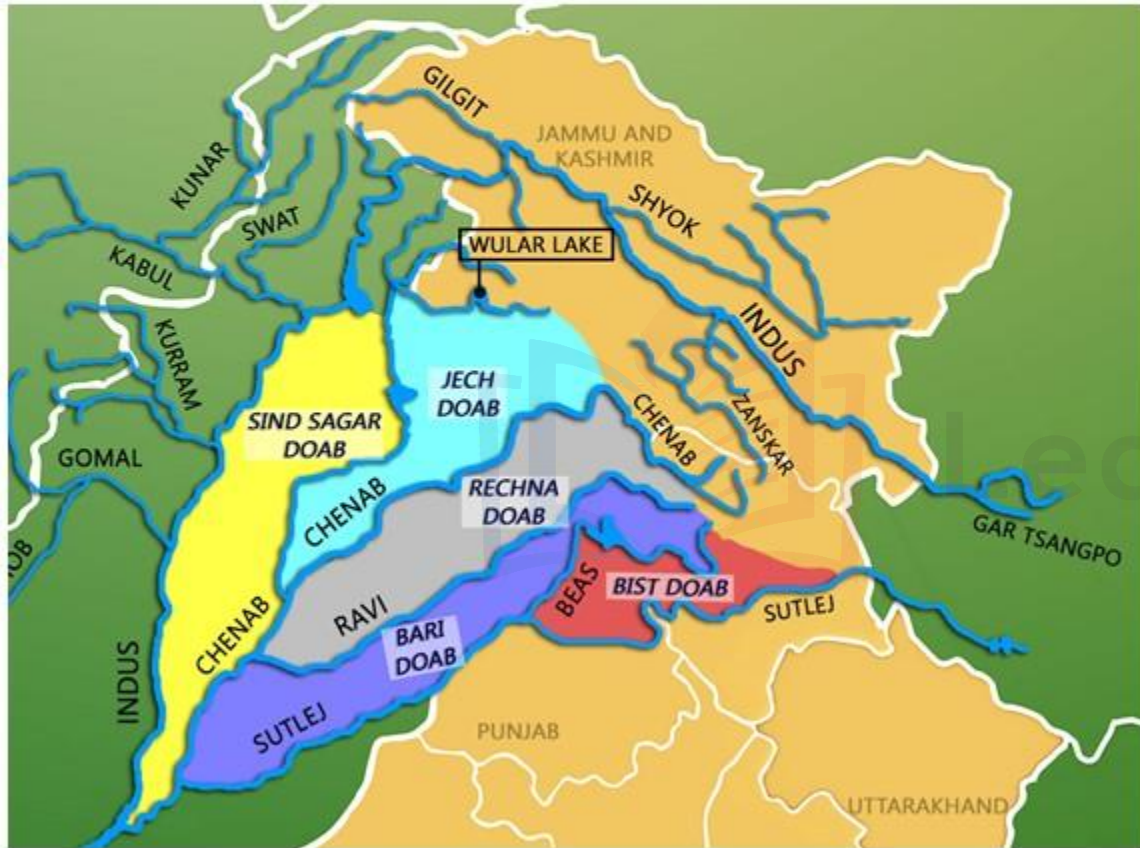
5 The depositional process by these rivers have united these doabs giving an homogenous appearance

6 The eastern boundary of the Punjab-Haryana Plain is marked by the Delhi Aravali ridge.

REGIONAL DIVISIONS OF GREAT PLAINS OF INDIA

2

PUNJAB-HARYANA PLAIN



7

From east-west, these doabs are as under:

- Bist-Jalandhar Doab**, lying between Beas and Sutlej
- Bari Doab** between the Beas and the Ravi
- Rechna Doab** between the Ravi and Chenab
- Chaj Doab** between the Chenab and the Jhelum
- Sind Sagar Doab** between the Jhelum-Chenab and the Indus

INDUS RIVER



QUESTION

Which of the following doabs are correctly matched?

- | | | |
|------------------------|---|-------------------|
| a) Bist Jalandhar Doab | - | Beas and Sutlej |
| b) Bari Doab | - | Beas and Ravi |
| c) Rechna Doab | - | Ravi and Chenab |
| d) Chej Doab | - | Chenab and Jhelum |
| e) Sindh Sagar Doab | - | Indus and Jhelum |

Which of the above is / are correct?

- | | |
|---------------|------------------------|
| a) a,b,d,e | b) b,c,d,e |
| c) a & d only | d) All of these |

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REGIONAL DIVISIONS OF GREAT PLAINS OF INDIA

2

PUNJAB-HARYANA PLAIN



Chos of Punjab overflowing

8 The northern part of this plain adjoining the Shiwalik hills has been intensively eroded by numerous streams called Chos. This has led to enormous gulying.

9 The erosion by the Chos is particularly noticed in Hoshiarpur district of Punjab.

10 In a short stretch of about 130 km nearly a hundred Chos debouch on the plains.

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2

PUNJAB-HARYANA PLAIN



11 The area between the **Ghaggar** and the **Yamuna Rivers** lies in Haryana and is often termed as “**Haryana Tract**”. It acts as water-divide between the Yamuna and Sutlej rivers.

12 The only river between Yamuna and Sutlej is the Ghaggar, which is considered to be the present day successor of the legendary Saraswati River.



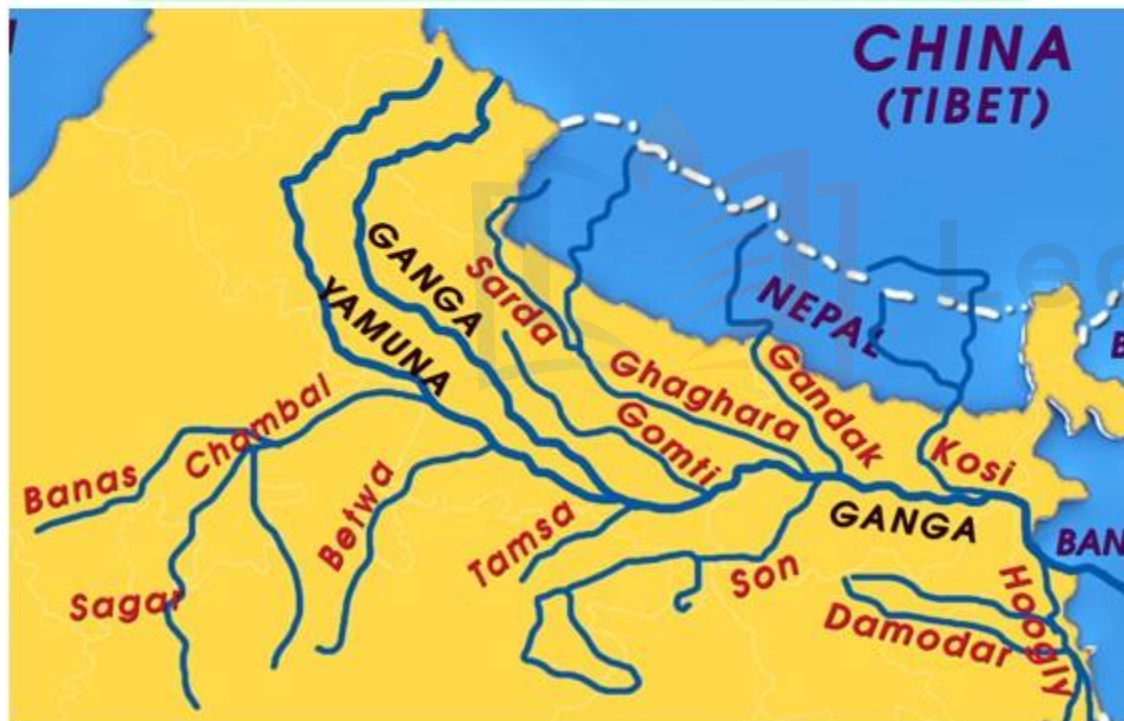
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REGIONAL DIVISIONS OF GREAT PLAINS OF INDIA

3

THE GANGA PLAIN

- 1 This is the largest unit of the Great Plain of India stretching from Delhi to Kolkata.



- 2 The Ganga along with its large number of tributaries originating in the Himalayas have brought large quantities of alluvium from the mountains and deposited it here building this extensive plain.
- 3 The peninsular rivers such as Chambal, Betwa, Ken, Son joining the Ganga river system have also contributed to the formation of this plain.
- 4 The general slope of the entire plain is to the east and the south-east.





PAKISTAN

The Northern
Mountain
Region

CHINA
(TIBET)

NEPAL

BHUTAN

Ganga
Plain

BANGLADESH

MYANMAR

I N D I A

INDIAN PHYSIOGRAPHY

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4

BRAHMAPUTRA PLAIN

- 1 This is also known as the **Assam Plain**, as most of its area is situated in Assam.



- 2 Though often treated as the eastern continuation of the Great plains of India, it is a well demarcated physical unit bordered by the eastern Himalayas of Arunachal Pradesh in the north, Patkai and Naga Hills in the east and the Garo, Kasi, Jaintia Hills in the south.

- 3 Its western boundary is formed by the Indo-Bangladesh border as well as the boundary of the lower Ganga plain.



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4

BRAHMAPUTRA PLAIN



4 It is an **aggradational plain** built by the depositional works of the Brahmaputra and its tributaries.

5 The innumerable tributaries of Brahmaputra river coming from the north **form a large number of alluvial fans.**

6 Consequently, the tributaries branch out in many channels giving birth to **river meandering leading to the formation of ox-bow lakes.**

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GANGA-BRAHMAPUTRA DELTA



- 1 This is the largest Delta in the World.
- 2 The Ganga river divides itself into several channels in the Delta area. The slope of the land here is, some mere 2 cm/km.
- 3 The seaward face of the Delta is started with large number of estuaries, mangrove swamps, islands etc.
- 4 Large part of the coastal delta is covered by tidal forests. These are called the Sundarbans, because of the predominance of Sundari tree here.

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SIGNIFICANCE OF THE PLAIN

- 1 With its fertile alluvial soils, flat surface, slow moving perennial rivers and favourable climate, the Great Plains of North India are of great significance.



- 2 It is the home of about half of the population of India, although it accounts for less than $1/4^{\text{th}}$ of the total area of the Country.
- 3 The Plain supports some of the highest population densities depending purely upon agro-based economy.
- 4 The extensive use of irrigation has made Punjab, Haryana and western parts of Uttar Pradesh, the granary of India (Prairies are called the granaries of the World).

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SIGNIFICANCE OF THE PLAIN

- 5** The entire Plain except the Thar Desert has a close network of roads and railways, which has led to large scale industrialisation and urbanization.



- 6** The development of trade and commerce in this Plain is a natural sequel of industrialisation and urbanization.

- 7** There are many religious places along the banks of the sacred rivers like the Ganga and Yamuna, which are very dearer to Hindus.

- 8** Here flourished the religions of Buddha and Mahavira and the movements of Bhakti and Sufism. In short, this vast Plain is the heart-throb of India.

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